

EBN122 SEMESTER TOETS/TEST 2 (17 OKT/OCT 2015)**ANTWOORDBLAD / ANSWER SHEET**

Van / Surname:	MENNO	Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans		English				

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (23)			
1.1	a) $I_0 = 0.57 \text{ A}$ ✓	b) $I_0 = 4 \text{ A}$ ✓	
1.2	a) $I_{0(5V)} = 0.833 \text{ A}$ ✓	b) $I_{0(2A)} = 2.67 \text{ A}$ ✓	
1.3	$R = 20 \Omega$ ✓	$I = (V_x/20 + 20) \text{ A}$ ✓	
1.4	$R_{Th} = 1.467 \Omega$ ✓	$V_{Th} = 4 \text{ V}$ ✓	
1.5	$R_{Th} = R_N =$ ✓		
1.6	$R_N = 1.33 \Omega$ ✓	$I_N = 0.5 \text{ A}$ ✓	
1.7	$R_L = 12 \Omega$ ✓		

VRAAG 2 – QUESTION 2 (22)			
2.1	$i_l = 5 \text{ mA}$ ✓	$v_l = -3 \text{ V}$ ✓	
2.2	$v_2 = -10 \text{ V}$ ✓	$v_s = 3 \text{ V}$ ✓	
2.3	a) $v_3 = -7 \text{ V}$ ✓	b) $v_3 = 15 \text{ V}$ ✓	
2.4	$v_4 = -4 \text{ V}$ ✓	$a = 1.33$ ✓	
2.5	$v_C(1s) = -4.32 \text{ V}$ ✓	$v_L(1s) = 2.71 \text{ mV}$ ✓	
2.6	$W_C = 2.5 \text{ mJ}$ ✓		